

Faculty of Applied Science and Technology**Lab #5 – Two- and Three-Dimensional Plots**

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November 5th, 2025**Procedures and results:**

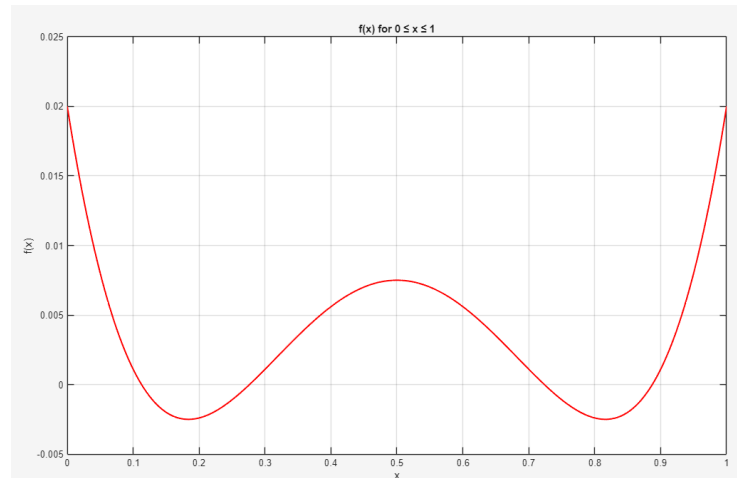
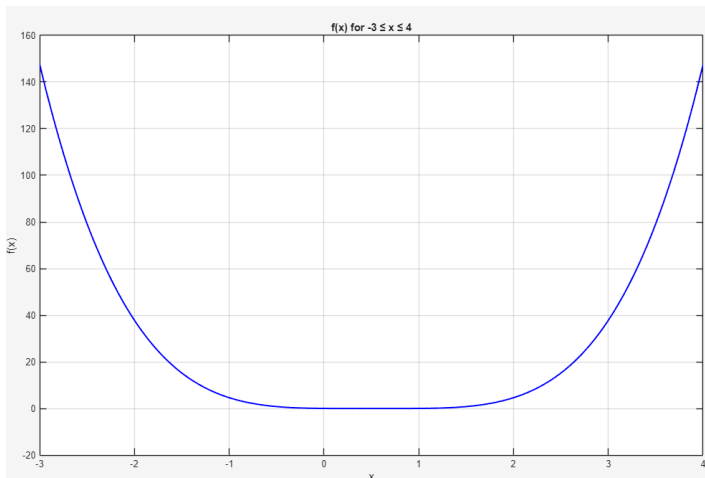
1.

1.1) Section 5: Question 5

```
% Question 5
% Define the equation
f = @(x) x.^4 - 2*x.^3 + 1.3*x.^2 - 0.3*x + 0.02;

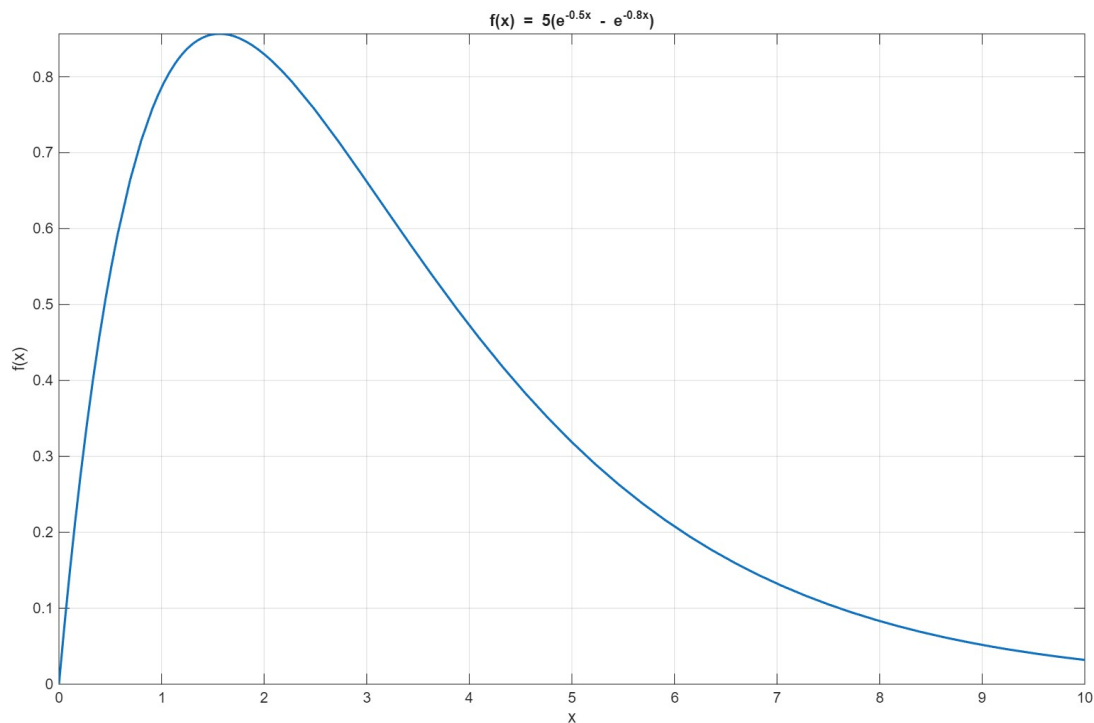
% Plot 1: for  $-3 \leq x \leq 4$ 
x1 = linspace(-3, 4, 500);
figure
plot(x1, f(x1), 'b', 'LineWidth', 1.5)
xlabel('x'); ylabel('f(x)'); grid on;
title('f(x) for  $-3 \leq x \leq 4$ ');

% Plot 2: for  $0 \leq x \leq 1$ 
x2 = linspace(0, 1, 500);
figure
plot(x2, f(x2), 'r', 'LineWidth', 1.5)
xlabel('x'); ylabel('f(x)'); grid on;
title('f(x) for  $0 \leq x \leq 1$ ');
```



1.2) Section 5: Question 6

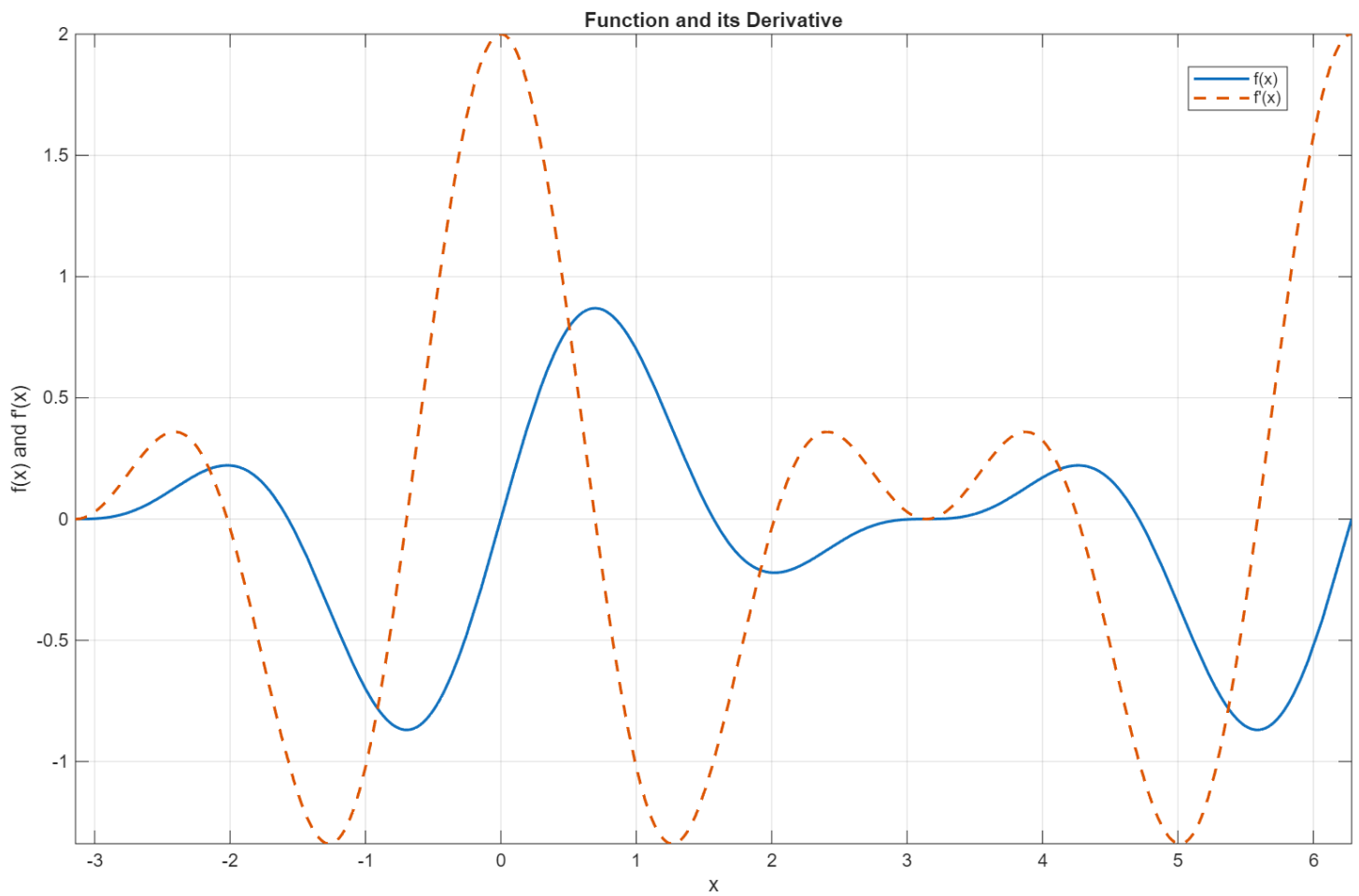
```
% Question 6
f = @(x) 5*(exp(-0.5*x) - exp(-0.8*x));
figure
fplot(f, [0 10], 'LineWidth', 1.5);
xlabel('x'); ylabel('f(x)'); grid on;
title('f(x) = 5(e^{-0.5x} - e^{-0.8x})');
```



1.3) Section 5: Question 7

```
% Question 7
syms x
f = sin(2*x) * cos(0.5*x)^2;
df = diff(f, x);

fplot(f, [-pi 2*pi], 'LineWidth', 1.5)
hold on
fplot(df, [-pi 2*pi], '--', 'LineWidth', 1.5)
xlabel('x'); ylabel('f(x) and f''(x)'); legend('f(x)', 'f''(x)');
title('Function and its Derivative'); grid on;
```

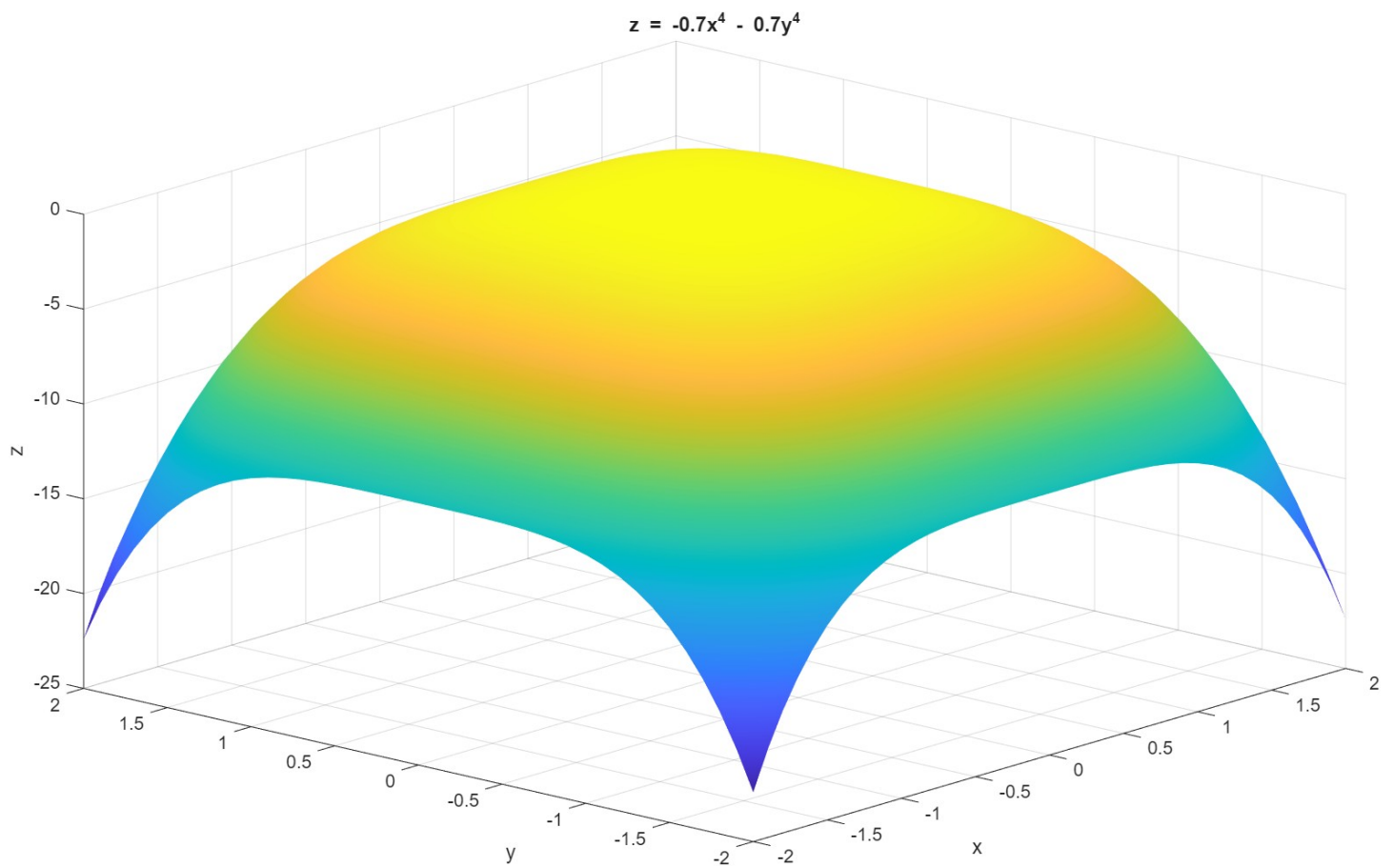


2) Section 10

2.1) Section 10: Question 5

```
% Question 5
[x, y] = meshgrid(-2:0.1:2, -2:0.1:2);
z = -0.7*x.^4 - 0.7*y.^4;

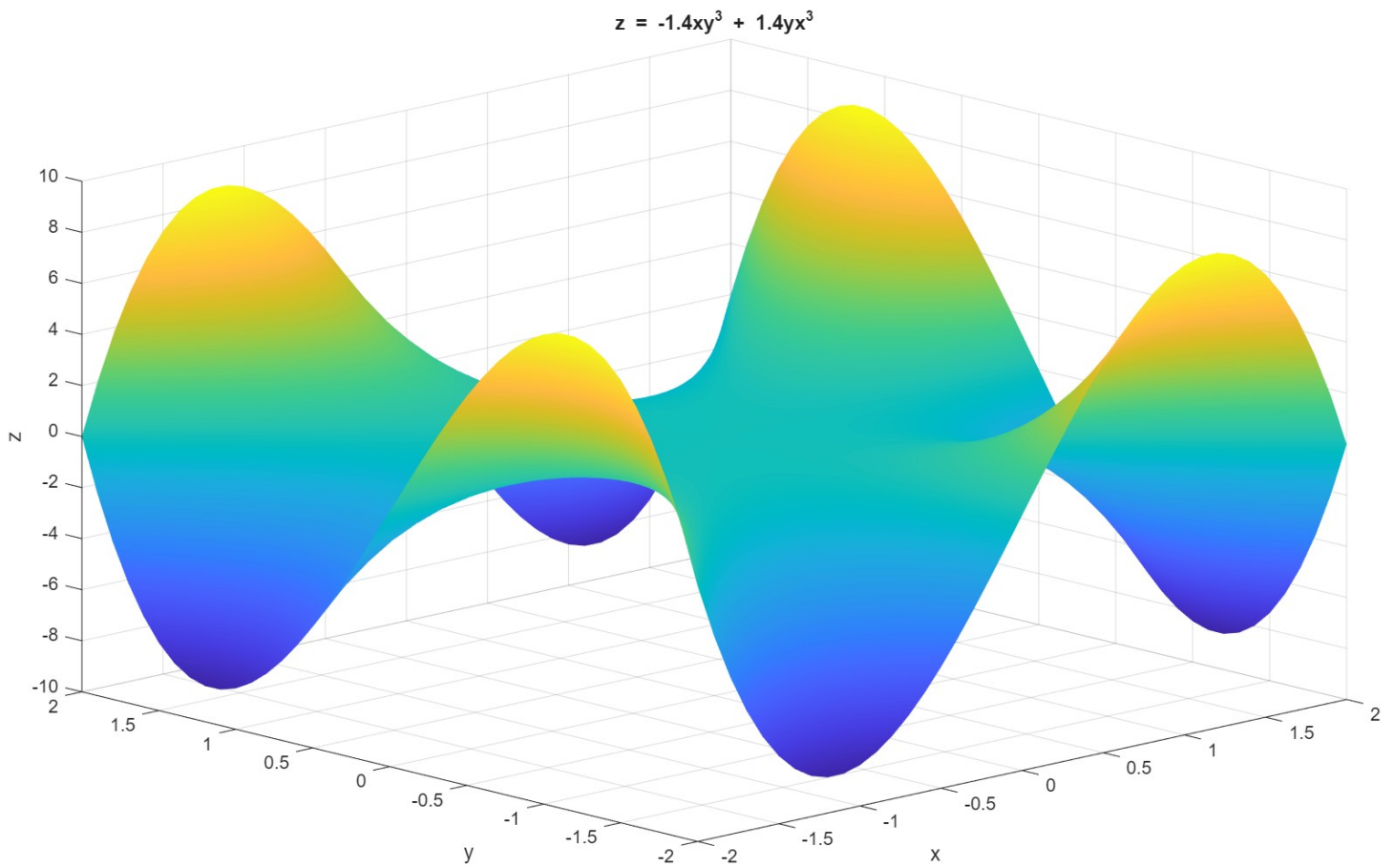
figure
surf(x, y, z)
xlabel('x'), ylabel('y'), zlabel('z');
title('z = -0.7x^4 - 0.7y^4')
shading interp
grid on
```



2.2) Section 10: Question 6

```
% Question 6
[x, y] = meshgrid(-2:0.1:2, -2:0.1:2);
z = -1.4*x.*y.^3 + 1.4*y.*x.^3;

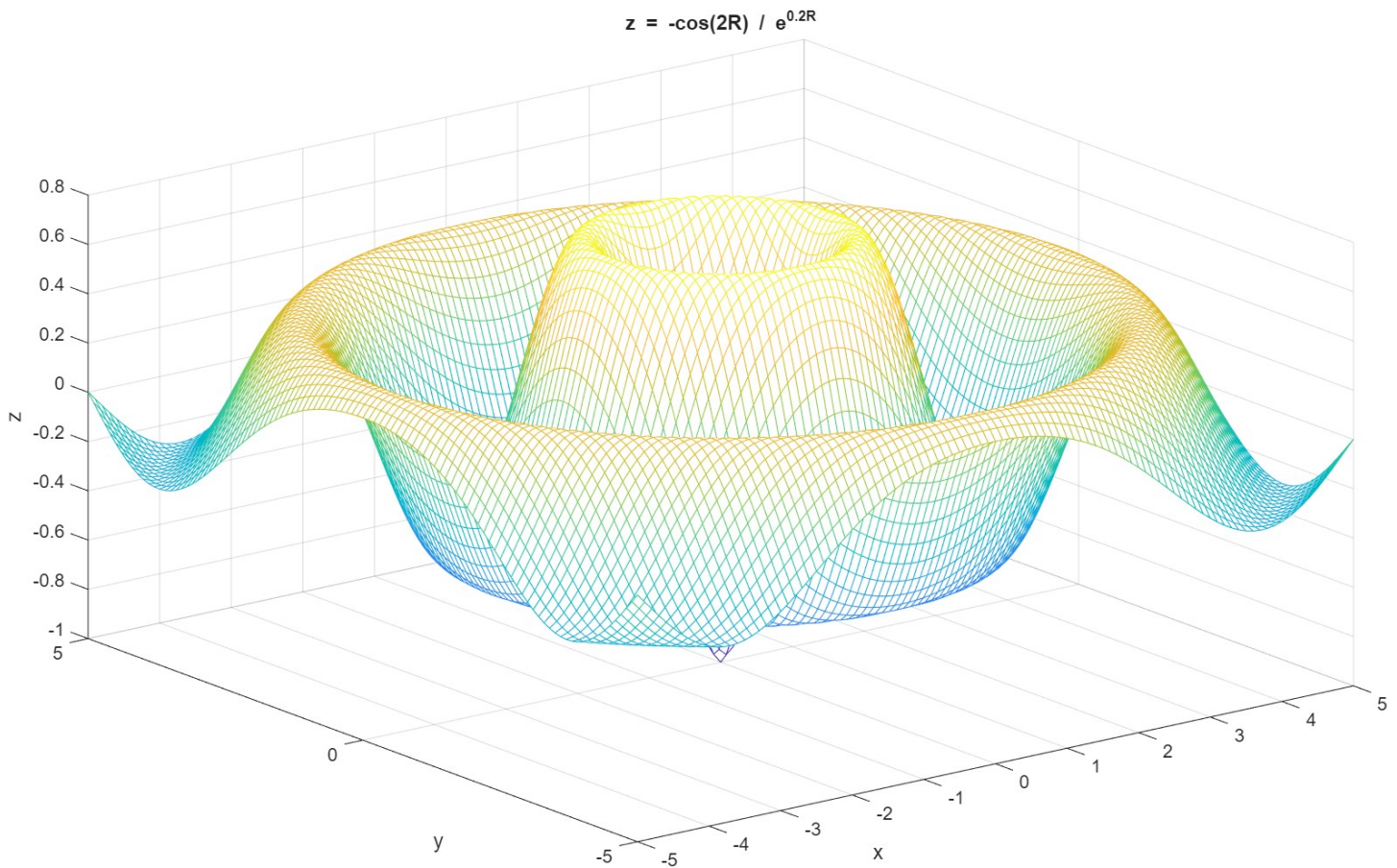
figure
surf(x, y, z);
xlabel('x'), ylabel('y'), zlabel('z');
title('z = -1.4xy^3 + 1.4yx^3');
shading interp; grid on;
```



2.3) Section 10: Question 7

```
% Question 7
[x, y] = meshgrid(-5:0.1:5, -5:0.1:5);
R = sqrt(x.^2 + y.^2);
z = -cos(2*R) ./ exp(0.2*R);

figure
mesh(x, y, z);
xlabel('x'), ylabel('y'), zlabel('z');
title('z = -cos(2R) / e^{0.2R}'); grid on;
```



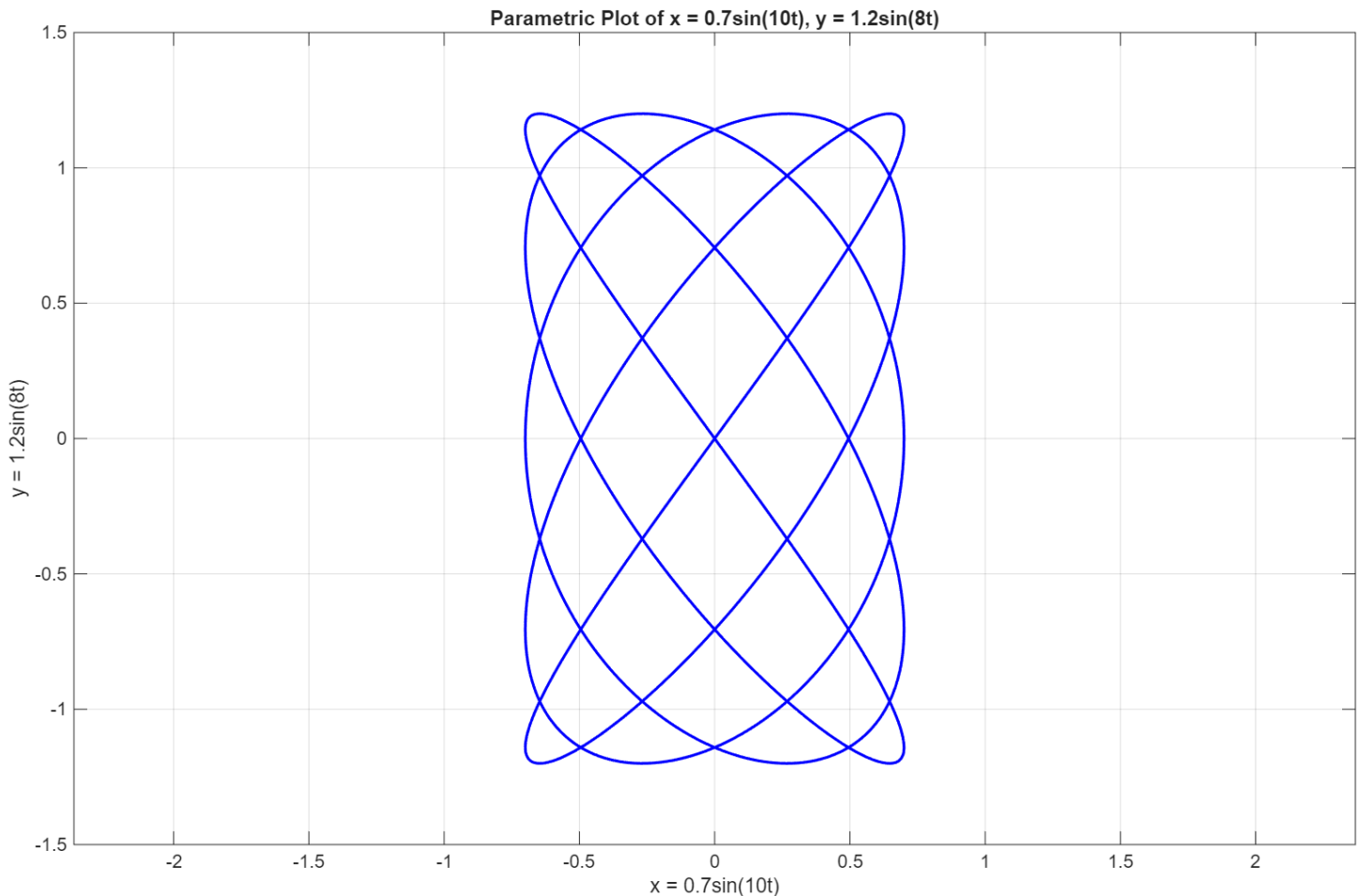
3) Section 5

3.1) Section 5: Question 9

```
% Question 9
% Define the parameter t
t = linspace(0, pi, 1000);

% Define the parametric equations
x = 0.7 * sin(10 * t);
y = 1.2 * sin(8 * t);

% Plot the curve
figure
plot(x, y, 'b', 'LineWidth', 1.5)
xlabel('x = 0.7sin(10t)'); ylabel('y = 1.2sin(8t)');
title('Parametric Plot of x = 0.7sin(10t), y = 1.2sin(8t)')
axis([-1.5 1.5 -1.5 1.5]); grid on; axis equal;
```



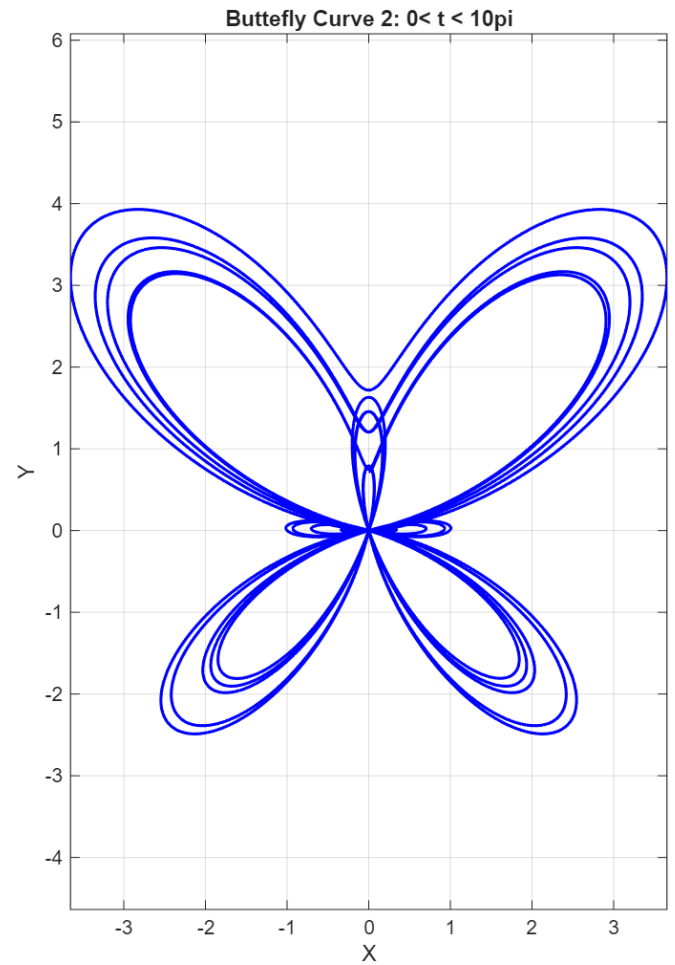
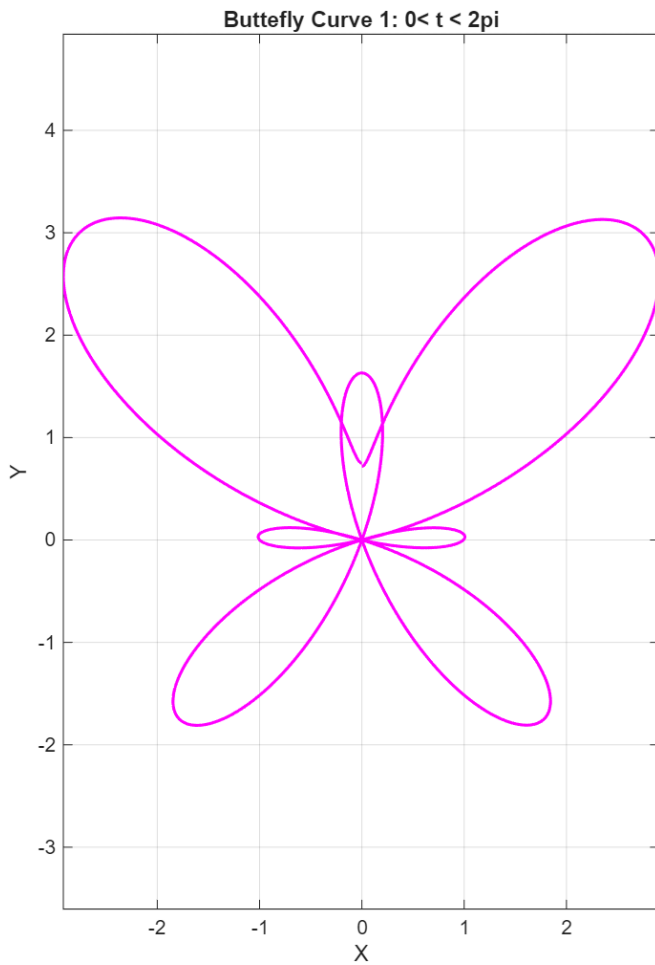
3.2) Section 5: Question 10

```
% Question 10
% Parameters for the graphs
t1 = linspace(0, 2*pi, 1000);
t2 = linspace(0, 10*pi, 5000);

% Define the parametric equations
x1 = sin(t1).*( exp(cos(t1)) - 2 .* cos(4.*t1) + sin(t1./12).^5);
y1 = cos(t1).*( exp(cos(t1)) - 2 .* cos(4.*t1) + sin(t1./12).^5);

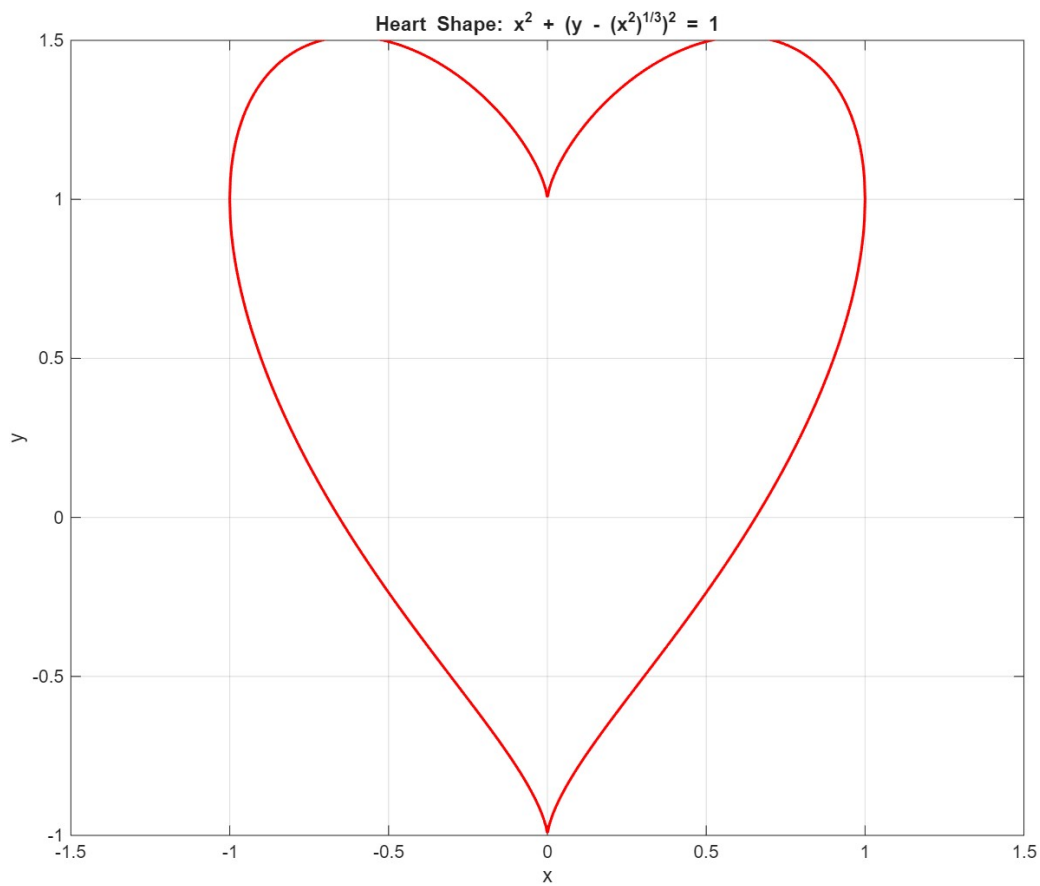
x2 = sin(t2).*( exp(cos(t2)) - 2 .* cos(4.*t2) + sin(t2./12).^5);
y2 = cos(t2).*( exp(cos(t2)) - 2 .* cos(4.*t2) + sin(t2./12).^5);

figure
% --- Plot 1: for  $0 < t < 2\pi$  ---
subplot(1,2,1)
plot(x1,y1,'m', 'LineWidth', 1.5); title('Butterfly Curve 1:  $0 < t < 2\pi$ ');
xlabel('X'); ylabel('Y'); axis equal; grid on;
% --- Plot 2: for  $0 < t < 10\pi$  ---
subplot(1,2,2)
plot(x2,y2,'b', 'LineWidth', 1.5); title('Butterfly Curve 2:  $0 < t < 10\pi$ ');
xlabel('X'); ylabel('Y'); axis equal; grid on;
```



3.3) Section 5: Question 15

```
% Question 15
% Parameters for the graphs
x = linspace(-1, 1, 1000);
|
% Define the parametric equations
y_upper = nthroot(x.^2, 3) + sqrt(1 - x.^2);
y_lower = nthroot(x.^2, 3) - sqrt(1 - x.^2);
% Plot the heart
figure
plot(x, y_upper, 'r', 'LineWidth', 1.5)
hold on
plot(x, y_lower, 'r', 'LineWidth', 1.5)
xlabel('x')
ylabel('y')
title('Heart Shape:  $x^2 + (y - (x^2)^{1/3})^2 = 1$ ')
axis equal
axis([-1.5 1.5 -1 1.5])
grid on
```

4) Section 10

4.1) Section 10: Question 17

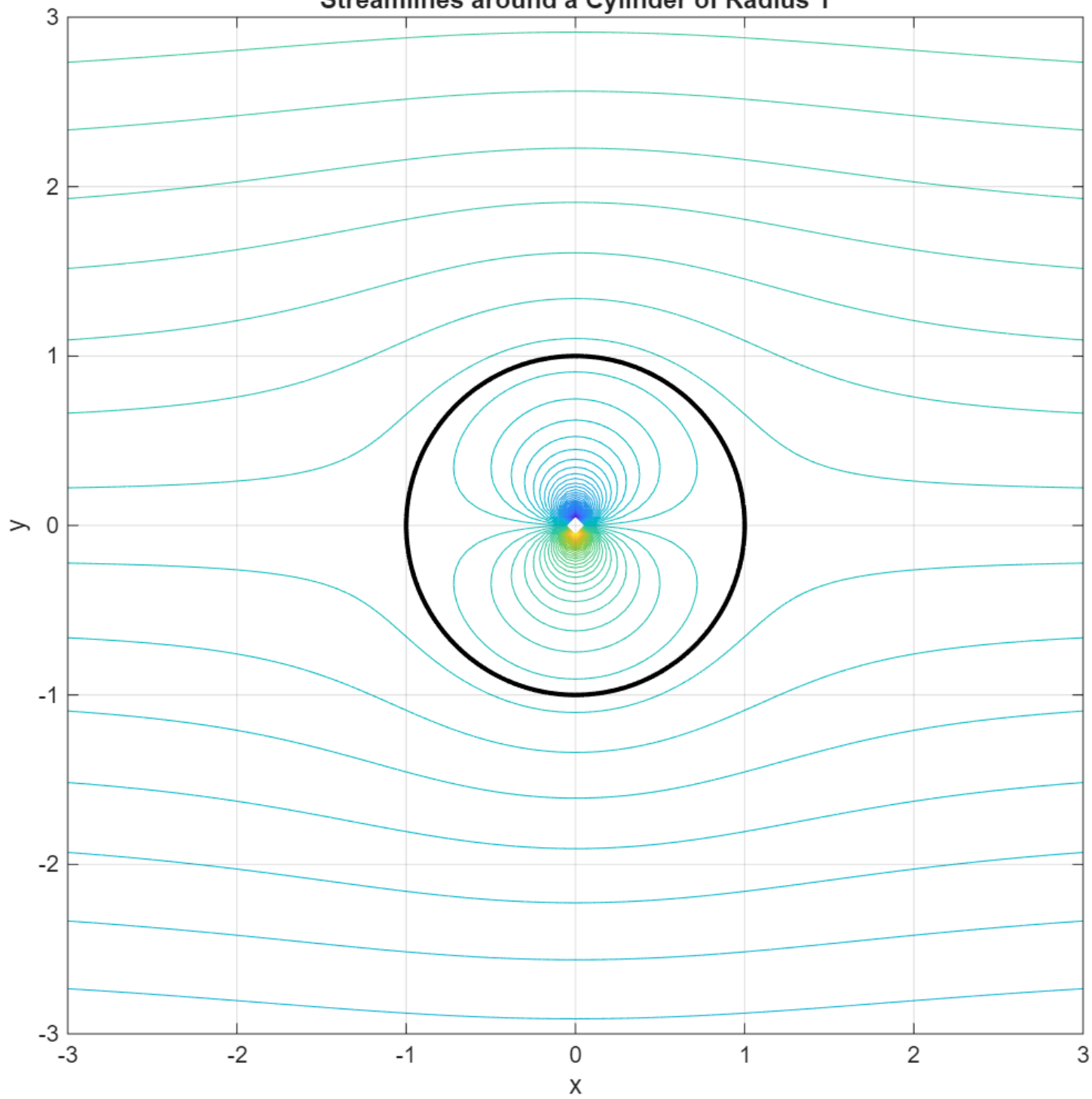
```
% Question 17
[x, y] = meshgrid(-3:0.05:3, -3:0.05:3);
psi = y - (y ./ (x.^2 + y.^2)); % Stream function

figure
contour(x, y, psi, 100)          % 100 contour levels
hold on

% Plot the cylinder of radius 1
theta = linspace(0, 2*pi, 200);
plot(cos(theta), sin(theta), 'k', 'LineWidth', 2)

xlabel('x'); ylabel('y');
title('Streamlines around a Cylinder of Radius 1');
axis equal; grid on;
```

Streamlines around a Cylinder of Radius 1



4.2) Section 10: Question 20

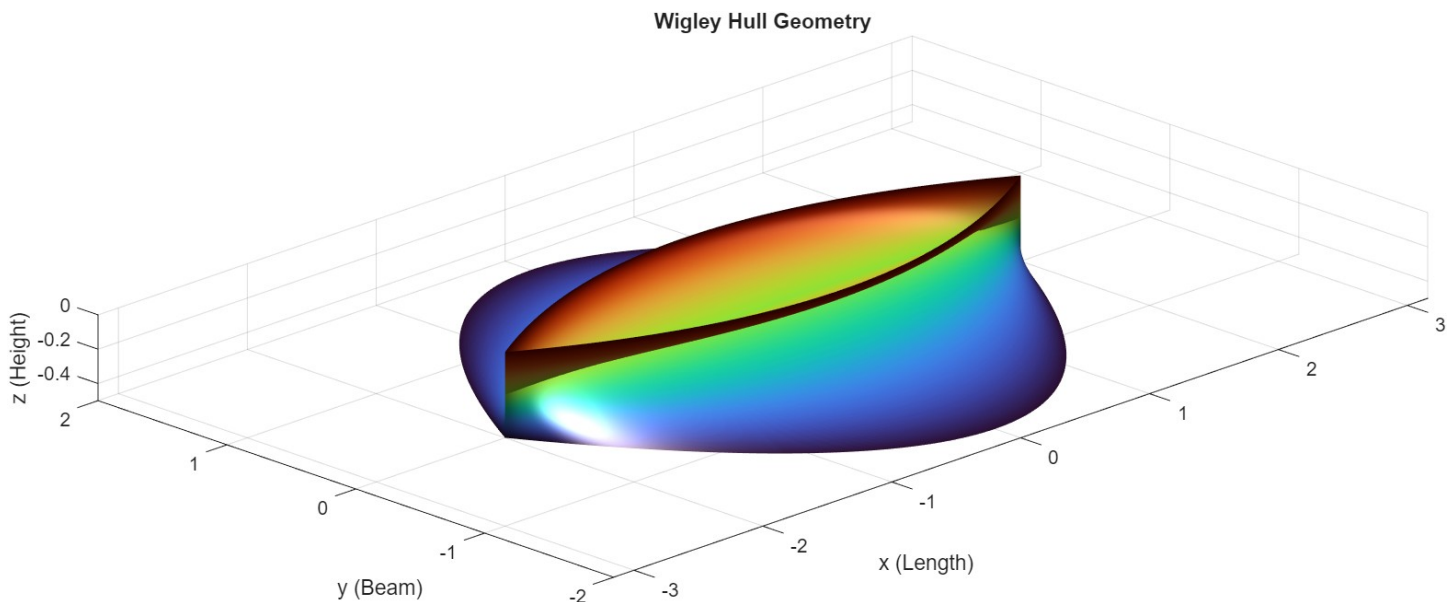
```
% Question 20
% Parameters
B = 1.2; % Beam (width)
L = 4;   % Length
T = 0.5; % Draft (height)

% Domain
[x, z] = meshgrid(linspace(-L/2, L/2, 200), linspace(-T, 0, 100));

% Wigley hull equation (half-breadth)
y = (B/2) .* (1 - (2.*x./L).^2) .* (1 - (2.*z./T).^2);

% Create figure
figure
hold on
surf(x, y, z, 'FaceColor', 'interp', 'EdgeColor', 'none'); % Starboard side
surf(x, -y, z, 'FaceColor', 'interp', 'EdgeColor', 'none'); % Port side
hold off

% Formatting
xlabel('x (Length)')
ylabel('y (Beam)')
zlabel('z (Height)')
title('Wigley Hull Geometry')
axis equal
grid on
colormap('turbo')
lighting gouraud
camlight headlight
view([-45 20])
```



4.3) Section 10: Question 21

```
% Question 21
% Mode II near-tip stresses (plot  $\sigma_{xx}$ ,  $\sigma_{yy}$ ,  $\sigma_{xy}$ )
KII = 300; % given K_II (units kept as-is)
r_min = 0.02;
r_max = 0.20;
nr = 120;
ntheta = 200;

r = linspace(r_min, r_max, nr);
theta = linspace(1e-4, pi/2 - 1e-4, ntheta); %  $0 < \theta < 90^\circ$ 

[R, TH] = meshgrid(r, theta); % TH in radians

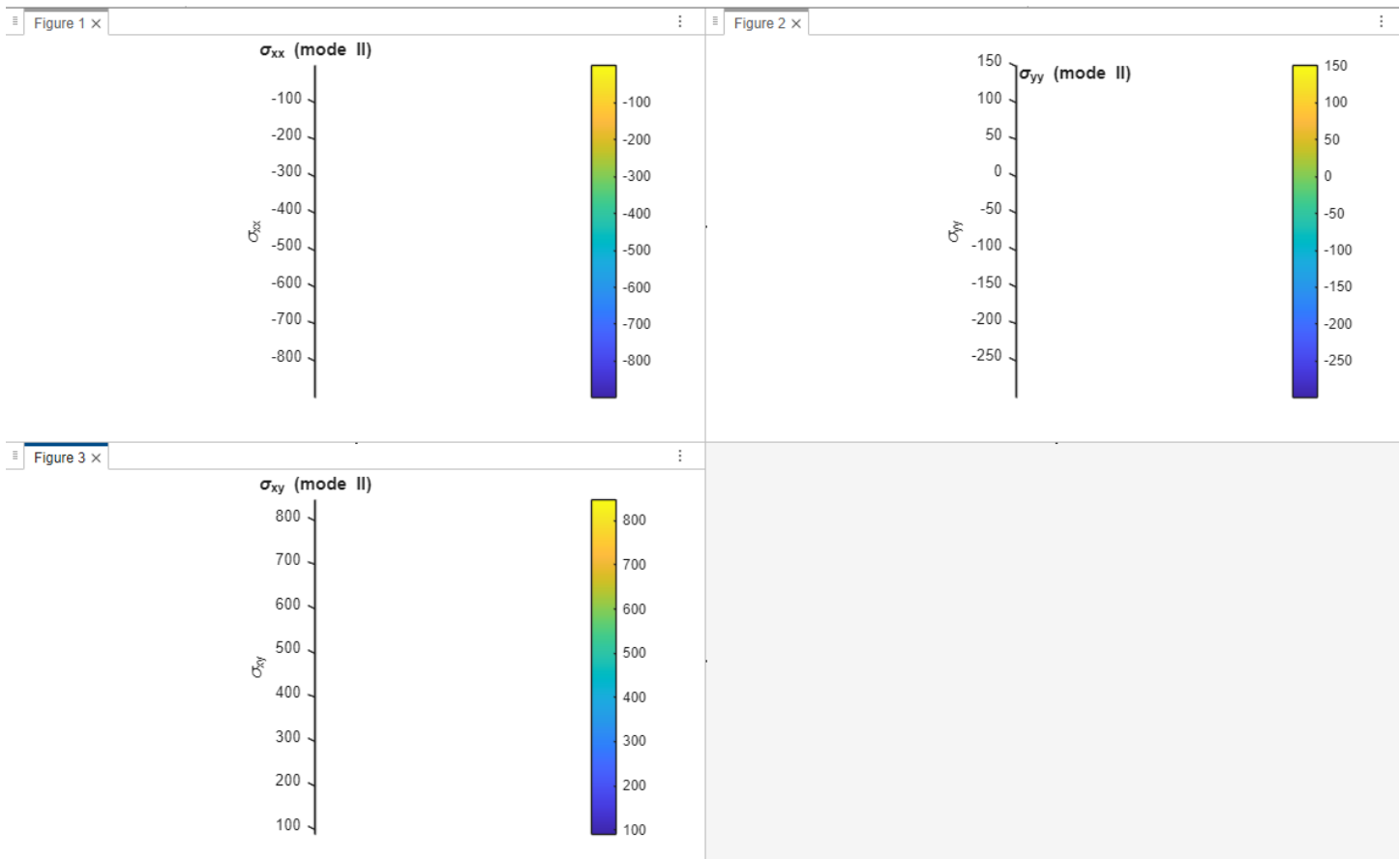
% common prefactor
pref = KII ./ sqrt(2*pi.*R);

% Stress expressions (as in the problem statement)
sigma_xx = - pref .* sin(TH/2) .* ( 2 + cos(TH/2) .* cos(3*TH/2) );
sigma_yy = pref .* sin(TH/2) .* ( cos(TH/2) .* cos(3*TH/2) );
sigma_xy = pref .* cos(TH/2) .* ( 1 - sin(TH/2) .* sin(3*TH/2) );

% Map to Cartesian coordinates for plotting
X = R .* cos(TH);
Y = R .* sin(TH);

% --- Plot  $\sigma_{xx}$  ---
figure('Color','w');
surf(X, Y, sigma_xx, 'EdgeColor','none');
shading interp
axis equal
xlabel('x'); ylabel('y'); zlabel('\sigma_{xx}');
title('\sigma_{xx} (mode II)');
colorbar
view(40,30)
grid on
|
% --- Plot  $\sigma_{yy}$  ---
figure('Color','w');
surf(X, Y, sigma_yy, 'EdgeColor','none');
shading interp
axis equal
xlabel('x'); ylabel('y'); zlabel('\sigma_{yy}');
title('\sigma_{yy} (mode II)', HorizontalAlignment='left', VerticalAlignment='top');
colorbar
view(40,30)
grid on

% --- Plot  $\sigma_{xy}$  ---
figure('Color','w');
surf(X, Y, sigma_xy, 'EdgeColor','none');
shading interp
axis equal
xlabel('x'); ylabel('y'); zlabel('\sigma_{xy}');
title('\sigma_{xy} (mode II)');
colorbar
view(40,30)
grid on
```



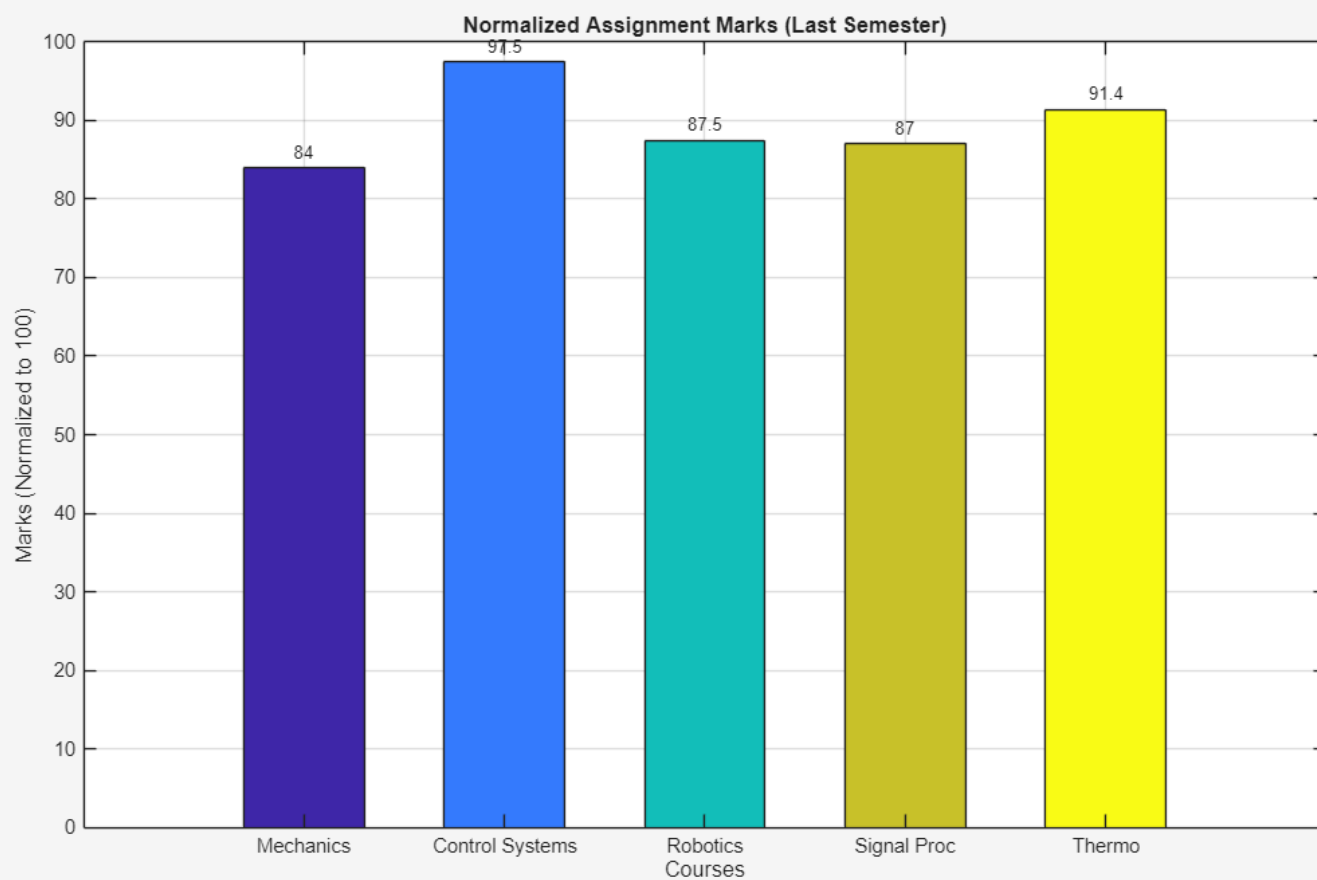
5) Assignment Marks Normalization and Plot

```
% Assignment Marks Normalization and Plot
% All last semesters assignment marks are locked so we will be assuming the
% data set.
% Course 1: Mechanics
% Course 2: Control Systems
% Course 3: Robotics
% Course 4: Signal Processing
% Course 5: Thermodynamics

% Original data
marks_obtained = [42, 78, 35, 87, 64]; % actual marks
max_marks      = [50, 80, 40, 100, 70]; % full marks
courses = {'Mechanics', 'Control Systems', 'Robotics', 'Signal Proc', 'Thermo'};

% Normalize to 100
normalized_marks = (marks_obtained ./ max_marks) * 100;

% Plot
figure;
b = bar(normalized_marks, 0.6);
b.FaceColor = 'flat';
b.CData = parula(5);
set(gca, 'XTickLabel', courses, 'FontSize', 11);
xlabel('Courses');
ylabel('Marks (Normalized to 100)');
title('Normalized Assignment Marks (Last Semester)');
ylim([0 100]);
text(1:5, normalized_marks + 2, string(round(normalized_marks,1)), HorizontalAlignment="center");
grid on;
```



References:

Gilat, A. (2017). MATLAB: An introduction with applications (6th ed.). John Wiley & Sons.